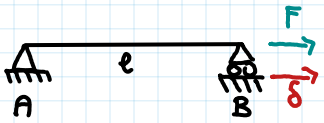
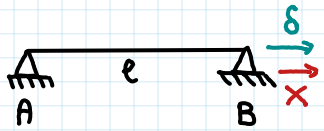
COEFFICIENTI ELASTICI di STRUTTURE ELEMENTARI

1



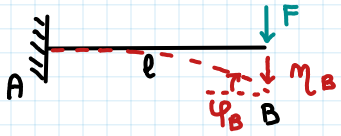
$$\delta_B = \frac{F\ell}{EA}$$

2



$$X = \frac{EA}{\ell}\,\delta$$

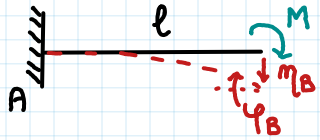
3



$$\varphi_B = -\frac{F\ell^2}{2EI}$$

$$\eta_B = \frac{F\ell^3}{3EI}$$

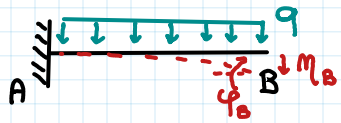
4



$$\varphi_B = -\frac{M\ell}{EI}$$

$$\eta_B = \frac{M\ell^2}{2EI}$$

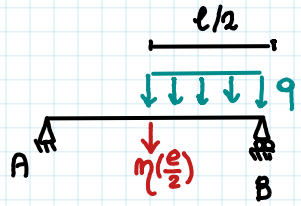
5



$$\varphi_B = -\frac{q\ell^3}{6EI}$$

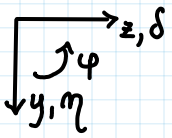
$$\eta_B = \frac{q\ell^4}{8EI}$$

6

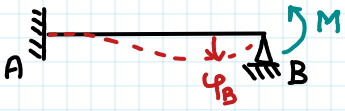


$$\eta\left(\frac{\ell}{2}\right) = \frac{5}{768}\,\frac{q\ell^4}{EI}$$

convenzione ⊕

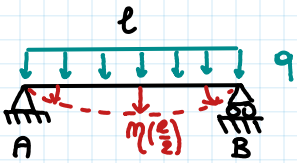


7



$$\varphi_B = \frac{M\ell}{4EI}$$

8

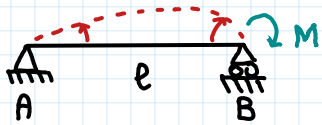


$$\varphi_A = -\frac{q\ell^3}{24EI}$$

$$\varphi_B = +\frac{q\ell^3}{24EI}$$

$$\eta\left(\frac{\ell}{2}\right) = \frac{5}{384}\,\frac{q\ell^4}{EI}$$

9

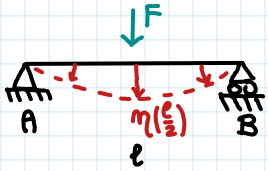


$$\varphi_A = +\frac{M\ell}{6EI}$$

$$\varphi_B = -\frac{M\ell}{3EI}$$

$$\eta\left(\frac{\ell}{2}\right) = -\frac{M\ell^2}{16EI}$$

10

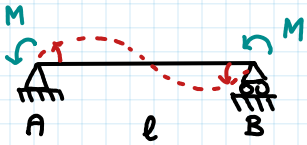


$$\varphi_A = -\frac{F\ell^2}{16EI}$$

$$\varphi_B = +\frac{F\ell^2}{16EI}$$

$$\eta\left(\frac{\ell}{2}\right) = \frac{F\ell^3}{48EI}$$

11

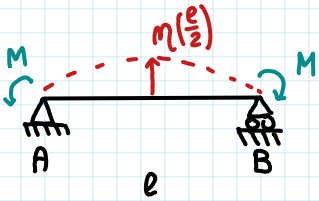


$$\varphi_A = \frac{M\ell}{6EI}$$

$$\varphi_B = \frac{M\ell}{6EI}$$

$$\eta\left(\frac{\ell}{2}\right) = 0$$

12



$$\varphi_A = \frac{M\ell}{2EI}$$

$$\varphi_B = -\frac{M\ell}{2EI}$$

$$\eta\left(\frac{\ell}{2}\right) = -\frac{M\ell^2}{8EI}$$